Northern University of Business and Technology, Khulna

Department of Computer Science and Engineering

CSE 4104 - Software Development III

Week 02 Submission: Project Proposal

Official Team Name	CSE4104-7F-T05
Section	7F
Project Title	AI Symptom Checker
Team Leader	Arnob Paul (ID: 230321390)
GitHub Repository	https://github.com/arnobpaul11/cse4104-7f-t05-ai-symptom-checker
Submission Date	5 November 2026

1. Project Title

AI Symptom Checker

A web-based health guidance tool that lets a user describe symptoms in plain Bangla or English, asks intelligent follow-up questions, and returns a structured result: possible causes to consider, a suggested urgency level, and general next steps. It is an information and guidance tool only. It does not give a medical diagnosis and does not replace a licensed doctor.

2. Problem Statement

What problem exists. When people feel unwell, many first search the internet or ask friends and family. The results are often generic, contradictory, or alarming. In smaller cities and rural areas of Bangladesh, reaching a doctor quickly can be difficult, so people may delay care for a serious condition, take the wrong medicine, or visit a crowded hospital for a minor problem.

Why it matters. Failing to recognise a warning sign such as chest pain, breathing difficulty, or sudden weakness can be dangerous. Unnecessary visits, on the other hand, cost time and money. People need a quick, clear answer to one question: *how urgent is this, and what should I do next?*

Why existing solutions are insufficient.

Existing system	Limitation
Doctronic, Ada Health	Built for general international users; not designed around local language and context; no simple shareable summary for a local doctor.
Infermedica, Buoy Health	Mostly fixed question flows and commercial products; limited free-text conversation.
WebMD, NHS symptom checkers	Rule-based decision trees with little personalisation and no memory of earlier conversations.
Semigran et al. (2015), BMJ	Audit study showing that the accuracy of online symptom checkers is limited and varies widely, so safe urgency guidance matters more than confident diagnosis.

Why AI is relevant. A fixed form cannot handle the many ways people describe how they feel, for example "burning feeling in my stomach after meals" or the same complaint written in Bangla. A language model can (1) understand free-text descriptions, (2) choose the next follow-up question based on earlier answers, like a clinician taking a history, (3) reason over symptoms together with age, existing conditions, and past history, and (4) explain the result in simple Bangla or English at any hour. Because language models can also make mistakes, our design does not trust the model alone: a separate rule-based safety layer checks for emergency warning signs, and every result carries a clear disclaimer.

3. Objectives

- Build a secure web application where registered users describe symptoms through a chat interface in Bangla or English.
- Use an AI model to ask adaptive follow-up questions instead of a fixed questionnaire.
- Produce a structured result: possible conditions to consider, urgency level (self-care, see a doctor soon, or emergency care), and general next steps.

- Implement a two-layer red-flag system (rule-based check plus AI check) that immediately recommends emergency care for serious warning signs.
- Keep a private symptom history and health profile (age, gender, existing conditions) and use them as context for later assessments.
- Generate a downloadable PDF summary that a user can show to a doctor.
- Provide an admin dashboard for usage statistics and for managing disclaimer and safety content.
- Measure AI quality by testing at least 30 prepared symptom scenarios and recording the results in the testing report.
- Deploy the system online and follow professional workflow: GitHub branches, meaningful commits, documentation.

4. Proposed Features

Features are grouped by priority so that the scope stays realistic for one 14-week semester. All Must-have features will be completed first; Should-have and Could-have features are attempted only after the core system is stable.

#	Feature	Description	Priority
F1	Authentication and roles	Register, login, JWT sessions, bcrypt password hashing, User and Admin roles with role-based access control.	Must
F2	AI chat symptom intake	Free-text symptom description with AI-generated adaptive follow-up questions.	Must
F3	Structured assessment	Possible causes with explanation, urgency level, and general next steps in structured JSON shown as a clear result card.	Must
F4	Red-flag emergency alert	Rule-based plus AI detection of serious warning signs; immediate emergency recommendation.	Must
F5	Profile-aware assessment	Age, gender, allergies, and existing conditions are included as context for the AI.	Must
F6	Symptom history	Private list of past conversations and results that the user can review or delete.	Must
F7	PDF health summary	Downloadable report of symptoms, answers, and result to share with a doctor.	Must
F8	Bangla-English support	Users type and read in Bangla or English; the AI replies in the chosen language.	Must
F9	Admin dashboard	User and assessment statistics; manage disclaimer text and safety content; manage users.	Must
F10	Feedback and AI monitoring	Thumbs up/down on responses; admin view of failures and feedback to improve prompts.	Should
F11	Personal health analytics	Charts of symptom frequency and history patterns.	Should
F12	Interactive body map	Select the body region where the symptom occurs.	Could
F13	Voice symptom input	Speech-to-text for describing symptoms.	Could
F14	Nearby facility finder	Locate nearby hospitals and clinics.	Could
F15	Audit logs and notifications	Record admin actions; reminders about saved reports.	Could

5. Technology Stack

Layer	Technology
Frontend	React.js, Tailwind CSS (responsive for desktop, tablet, and mobile); Recharts for charts
Backend	Node.js + Express.js; REST APIs; JWT authentication; bcrypt; input validation
Database	PostgreSQL hosted on Supabase
AI Integration	Google Gemini API (primary) or OpenAI API, accessed only from the backend through a provider-independent service module so the model can be swapped; prompt engineering with structured JSON output
PDF generation	A Node.js PDF library (for example PDFKit) on the backend
Deployment	Vercel (frontend), Render (backend), Supabase (database)

Layer	Technology
Collaboration and testing	GitHub (branches: main, development, frontend, backend, ai-integration), Figma (UI design), Postman (API testing), draw.io and dbdiagram.io (diagrams)

AI workflow

User message → backend rule-based red-flag check → prompt built from message, profile, and history → AI model → backend validates the JSON response → result saved to database → shown in the frontend. API keys and database credentials are stored only in environment variables and are never committed to GitHub.

6. Expected Outcome

What users gain. A fast, understandable answer about how urgent their symptoms are, in their own language, available at any hour, with a printable summary that makes a doctor visit more efficient.

How the workflow improves. Instead of searching many websites and reading conflicting results, the user has one guided conversation that remembers their profile and history and ends with a clear next step.

Benefits of AI integration. Natural-language understanding, adaptive questioning, personalised reasoning, and plain-language explanation, which a fixed questionnaire cannot provide.

Project outcome. A deployed, tested, and documented full-stack AI system with a measured AI evaluation, suitable as a portfolio project.

7. Feasibility, Risks, and Safety

Risk	How we handle it
AI gives wrong or unsafe advice	No diagnosis wording; urgency-focused output; rule-based red-flag layer independent of the AI; mandatory disclaimer on every result; scenario testing in Week 10.
Health data is sensitive	Password hashing, HTTPS, per-user data access, users can delete their history, only test data (no real patient data) during development.
AI API quota or cost limits	Provider-independent AI module, short prompts, rate limiting, and a fallback provider.
Scope too large	Priority levels (Must, Should, Could); Could-have features are dropped first if time is short.
Uneven team contribution	Clear roles, branch-based workflow, weekly commits by every member, weekly progress review by the team leader.

Scope note. The system is not a medical device. It provides general information and urgency guidance only, and always advises users to consult a licensed doctor.

8. References

- Semigran, H. L., Linder, J. A., Gidengil, C., and Mehrotra, A. (2015). Evaluation of symptom checkers for self diagnosis and triage: audit study. *BMJ*, 351, h3480.
- Reviewed systems: Doctronic, Ada Health, Infermedica, Buoy Health, WebMD symptom checker, NHS online symptom checker.